THE HOM FUNCTOR OF FINITE TYPE MODULES OVER A NOETHERIAN BASE IS AFFINE OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-hom (canonical tag 08JT). cited at statement line 808. Standing situation of the Hom section.
- Definition spaces-perfect-definition-approximation-holds (canonical tag 08HI). cited at proof lines 826-827. Definition of the approximation.
- Theorem spaces-perfect-theorem-approximation (canonical tag 08HP). cited at proof line 828. Existence of the perfect approximation P.
- Remark sites-cohomology-remark-before-Leray (canonical tag 08J6). cited at proof lines 852-853. Leray rewriting of the functor.
- Lemma spaces-perfect-lemma-base-change-RHom (canonical tag 08JQ). cited at proof lines 860-861. Base change of Rf_*RSheafHom.
- Lemma spaces-perfect-lemma-ext-perfect (canonical tag 0DKK). cited at proof lines 867-868. K is perfect.
- Lemma lemma-cohomology-perfect-complex (canonical tag 08JL). cited at proof lines 870-871. Representability applied to K.

Source reference. Original definition 08JT: sources/stacks-chow/tags.txt, source lines 10861–10861. Cited at statement line 808: the lemma is phrased in this situation; load-bearing.

Source reference. Original definition 08HI: sources/stacks-chow/tags.txt, source lines 10781–10781. Cited at proof lines 826-827: the approximation P -> F of the triple (X, F, -1); load-bearing.

Source reference. Original theorem 08HP: sources/stacks-chow/tags.txt, source lines 10787–10787. Cited at proof line 828: existence of the approximation P; load-bearing.

Source reference. Original remark 08J6: sources/stacks-chow/tags.txt, source lines 10839–10839. Cited at proof lines 852-853: Hom over X_T computed as H^0 of RSheafHom and pushed to T; load-bearing.

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Source reference. Original lemma 08JQ: sources/stacks-chow/tags.txt, source lines 10858–10858. Cited at proof lines 860-861: Rf_{T,*}RSheafHom(P_T, G_T) = Lg^* Rf_*RSheafHom(P, G); load-bearing.

Source reference. Original lemma 0DKK: sources/stacks-chow/tags.txt, source lines 17015–17015. Cited at proof lines 867-868: K is perfect; load-bearing.

Source reference. Original lemma 08JL: sources/stacks-chow/tags.txt, source lines 10854–10854. Cited at proof lines 870-871: applied to K with negative cohomology vanishing to conclude; load-bearing.

Lemma 0.2. *In Situation 08JT assume that*

(1) *B is a Noetherian algebraic space,*
(2) *f is locally of finite type and quasi-separated,*
(3) *$\mathcal{F}$ is a finite type $\mathcal{O}_X$-module, and*
(4) *$\mathcal{G}$ is a finite type $\mathcal{O}_X$-module, flat over B, with support proper over B.*

Then the functor $Hom(\mathcal{F}, \mathcal{G})$ is an algebraic space affine and of finite presentation over B.

Proof. We may replace X by a quasi-compact open neighbourhood of the support of $\mathcal{G}$, hence we may assume X is Noetherian. In this case X and f are quasi-compact and quasi-separated. Choose an approximation $P \to \mathcal{F}$ by a perfect complex P of the triple $(X, \mathcal{F}, -1)$, see Derived Categories of Spaces, Definition 08HI and Theorem 08HP). Then the induced map

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Hom}_{D(\mathcal{O}_X)}(P, \mathcal{G})$$

is an isomorphism because $P \to \mathcal{F}$ induces an isomorphism $H^0(P) \to \mathcal{F}$ and $H^i(P) = 0$ for $i > 0$. Moreover, for any morphism $g : T \to B$ denote $h : X_T = T \times_B X \to X$ the projection and set $P_T = Lh^*P$. Then it is equally true that

$$\mathrm{Hom}_{\mathcal{O}_{X_T}}(\mathcal{F}_T, \mathcal{G}_T) \longrightarrow \mathrm{Hom}_{D(\mathcal{O}_{X_T})}(P_T, \mathcal{G}_T)$$

is an isomorphism, as $P_T = Lh^*P \to Lh^*\mathcal{F} \to \mathcal{F}_T$ induces an isomorphism $H^0(P_T) \to \mathcal{F}_T$ (because h^* is right exact and $H^i(P) = 0$ for $i > 0$). Thus it suffices to prove the result for the functor

$$T \longmapsto \mathrm{Hom}_{D(\mathcal{O}_{X_T})}(P_T, \mathcal{G}_T).$$

By the Leray spectral sequence (see Cohomology on Sites, Remark 08J6) we have

$$\mathrm{Hom}_{D(\mathcal{O}_{X_T})}(P_T, \mathcal{G}_T) = H^0(X_T, R\mathcal{H}om(P_T, \mathcal{G}_T)) = H^0(T, Rf_{T,*}R\mathcal{H}om(P_T, \mathcal{G}_T))$$

where $f_T : X_T \to T$ is the base change of f. By Derived Categories of Spaces, Lemma 08JQ we have

$$Rf_{T,*}R\mathcal{H}om(P_T, \mathcal{G}_T) = Lg^*Rf_*R\mathcal{H}om(P, \mathcal{G}).$$

By Derived Categories of Spaces, Lemma 0DKK the object $K = Rf_*R\mathcal{H}om(P, \mathcal{G})$ of $D(\mathcal{O}_B)$ is perfect. This means we can apply Lemma 08JL as long as we can prove that the cohomology sheaf $H^i(Lg^*K)$ is 0 for all $i < 0$ and $g : T \to B$ as above. This is clear from the last displayed formula as the cohomology sheaves of $Rf_{T,*}R\mathcal{H}om(P_T, \mathcal{G}_T)$ are zero in negative degrees due to the fact that $R\mathcal{H}om(P_T, \mathcal{G}_T)$ has vanishing cohomology sheaves in negative degrees as P_T is perfect with vanishing cohomology sheaves in positive degrees. □